	AzureSky	
	ISO Volatile matter determination	Document no: AS006IW
	Compiled by:	Sifiso Nyambi - Laboratory Manager
	Approved by:	Nonhlanhla Msibi - Operations Manager

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1. INTRODUCTION

The volatile matter content of coal is determined as the loss in mass, less that due to moisture when coal is incinerated in specified conditions in the absence of air.

Use form (AS013TF) to record the furnace temperature.

2. SCOPE AND FIELD OF APPLICATION

This test method describes the procedure for the determination of the volatile matter in hard coal samples.
The results are verified using certified reference material and by plotting a control chart. The chart contains upper and lower limits, any result that falls outside this range is said to be “out or failed”. Using this chart, the lab can detect problems with either the equipment, staff or possible environmental effects. In essence, the chart is used to initiate an investigation by the analyst to determine the probable cause.

3. REFERENCES

ISO 562:2010 - Hard coal and coke - Determination of volatile matter

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NOTE: ONLY PERSONNEL WHO ARE DECLARED COMPETENT ARE TO PERFROM LABORATORY ACTIVITIES. PERSONNEL NOT DECLARED ARE TO STRICTLY WORK UNDER SUPERVISION OF A LAB MANAGER/SUPERVISOR OR COMPETENT PERSONNEL.

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4. EQUIPMENT

4.1. Electrically heated muffle furnace in which an adequate zone, at a constant and uniform temperature of 900 ±5°C can be maintained. The heat capacity shall be such that with an initial temperature of 900°C, a minimum temperature of 896°C is regained within four minutes after the insertion of a cold crucible stand and its crucibles into the furnace.

4.2 Cylindrical crucible with lid, both of glazed silica, 38mm height x 25mm O.D., conforming to ISO 562:2010.

Verification of furnaces is to be done every Monday of every week. Insert the probe into the furnace; Ash furnace readings are taken as follows: 1st reading at 30 minutes, 2nd reading at 45 minutes and last reading at 1 hour. For volatile furnace, 1st read at 4 minutes after insertion of the external probe and the second and third read at 7-minute intervals consecutively. Apply the correction factors as specified in the calibration certificates of the external thermometers. The principle is to mathematically add the correction factor. This is to be documented in the AS024TF form.

NOTE: The fit of the lid is critical and shall be selected so that when placed in position, the horizontal clearance around the lip of the crucible is not more than 0.5mm. After selection, each crucible and lid shall be ground together by means of grinding paste to give smooth, well-fitting surfaces and then identified with a distinguishing number on both the crucible and lid using heat resistant paint/ an electrical engraver.

New crucibles together with the lids shall be verified for mass. Crucibles together with the lids shall be between 10g to 14g. Crucibles not meeting this requirement shall not be used. Crucibles together with their lids shall be labelled to prevent interchanging of the lids which could affect the crucible mass.

4.3. Heat resistant wire stands capable of suspending four crucibles.

4.4. Analytical balance accurate to 0.0001g

4.5. Furnace tongs capable of rapidly inserting or removing crucible stands with crucibles from the furnace.

4.6 Stopwatch or timer capable of accurately timing seven minutes ±7 seconds.

4.7 A desiccator.

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4.8 Desiccant - self indicating silica gel.

Note: Ensure all equipment and materials used are verified against factory specifications and logged in the equipment register prior to use. Calibrations (where necessary) and verification of equipment are to be attended prior to use. Calibration services shall only be sourced from ISO/IEC 17025 accredited labs. Verification certificates from manufacturers are accepted for new equipment provided to address traceability.(ISO 17025; Clause 6.4.)

5 INSTRUCTIONS

5.1 Collect conditioned and bottled samples on which volatile matter content shall be determined.

5.2 Make sure the crucibles are clean and in numerical order.

5.3 Tare the balance, weigh the crucibles with lids and record weights (M1) onto the crucible field on the volatile matter analysis sheet (LIMS).

5.4 Tare the balance, mix the sample thoroughly with a spatula and introduce between 0.95g and 1.05g of sample into the crucibles.

5.5 Tap the crucibles lightly on a hard top, weigh and record onto the crucible plus sample field on the volatile matter analysis sheet.

5.6 Place crucibles in stands and proceed to the furnace room.

5.7 With the crucibles placed in stands and being satisfied that the furnace temperature is stable at 900°C, place the stand with the crucibles into the furnace so that the thermocouple tip is at the center of the four crucibles. This operation shall be done swiftly so as to prevent unnecessary temperature loss. Minimize the time taken to open, insert and close the furnace. If less than four determinations are carried out, any vacant places in the stand should be filled with empty crucibles.

5.8 Record the initial temperature on the volatile temperature recording form (AS013TF)

5.9 Insert the crucibles into the volatile furnace in the hot zone, close the furnace door immediately and start the stopwatch.

5.10 Record the temperature displayed after 4minutes to check if it has recovered to 896˚C.

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(*If the furnace has not recovered, reject the results and immediately run repeat analyses*.)

5.11 Keep the crucible in the furnace for 7min + 2 seconds and record the end temperature.

5.12 Remove the crucible and place on a metal plate for 5min.

5.13 Proceed to the conditioning room and place them in the desiccator for 10min.

5.14 Proceed to the analytical room, tare the balance weigh back and record weight (M3) onto the final mass field on the volatile matter analysis sheet.

5.15 Run control sample after every 4 samples, this will allow you to check for possible drift on the instrument.

SAFETY NOTE: WHEN THE STAND AND CRUCIBLES ARE REMOVED FROM THE FURNACE IT IS EXTREMELY HOT. ALWAYS USE TONGS WHEN HANDLING THIS HOT APPARATUS.

5.16 If the control sample is out of repeatability limit, investigate the root cause and repeat the control sample.

5.17 If the control sample is within repeatability, calculate the results to dry basis and plot the control chart.

5.18 Clean the crucibles by incineration in a furnace for 5 minutes after each test.

6 CALCULATIONS

Note: Calculation is done manually as below or using the Prox calculation spreadsheet.

Mass % Volatile = $(\frac{m2-m3}{m2-m1} * 100) - IM$

Where: M1 = Mass of empty crucible in gram.

M2 = Mass of crucible with conditioned coal in gram.

M3 = Mass of crucible with burnt coal in gram.

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7 RECORDS

Records of all analysis, calculations and reports shall be stored in the files for that purpose or stored on the database.

8 REPEATABILITY AND REPRODUCIBILITY

The repeatability between duplicate results of the same sample of coal with less than 10% volatile matter is 0.3% and 3% of the average results for volatiles higher than 10%.

The reproducibility of volatile matter results (dry basis) is 0.5% absolute or 4% of the mean results whichever is greater.

Note: The final result shall be reported to the nearest 0.1%

9 FURNACE VERIFICATIONS

9.1. It is very important that the temperature of the furnace is frequently verified to ensure that we accurately determine the matter that is liberated at the specified temperature as per procedure.

9.2. Volatile furnaces shall be verified every other Monday (weekly).

The frequency can however change depending on the collective data.

9.3. At the start of the verification, the furnace temperature should be 900 ± 5 ºC

9.4 The temperature after 4 minutes should be verified as the furnace should the 896°C temperature in 4 minutes.

9.5 Specify the probe correction factor and add it from the probe reading e.g. if the probe

correction factor is 3°C and the probe reading is 907°C then the corrected probe reading will be 910°C. And if the probe correction is -3°C and the probe reading is 905°C, then the corrected probe reading is 902°C.

9.6 Report to the supervisor if the results are out of the tolerance as per section 4.1

10. RESPONSIBILITIES

The Analyst shall be responsible for plotting the control charts and carrying out the necessary repeats. The analyst shall be responsible for initiating any investigation on deviations from this procedure. The supervisor shall ensure that the analyst is trained and deemed competent on this method. The supervisor shall arrange for calibrations as per calibration schedule,

11. REVISION LIST

Revision no	Content changed	Reason for revision
01	No change, only addition of information	Under the equipment section, the process of verification of furnaces added as an improvement to the system.

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